



جامعة طرابلس | كلية طب الأسنان

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# DEVELOPMENT OF THE FACE.

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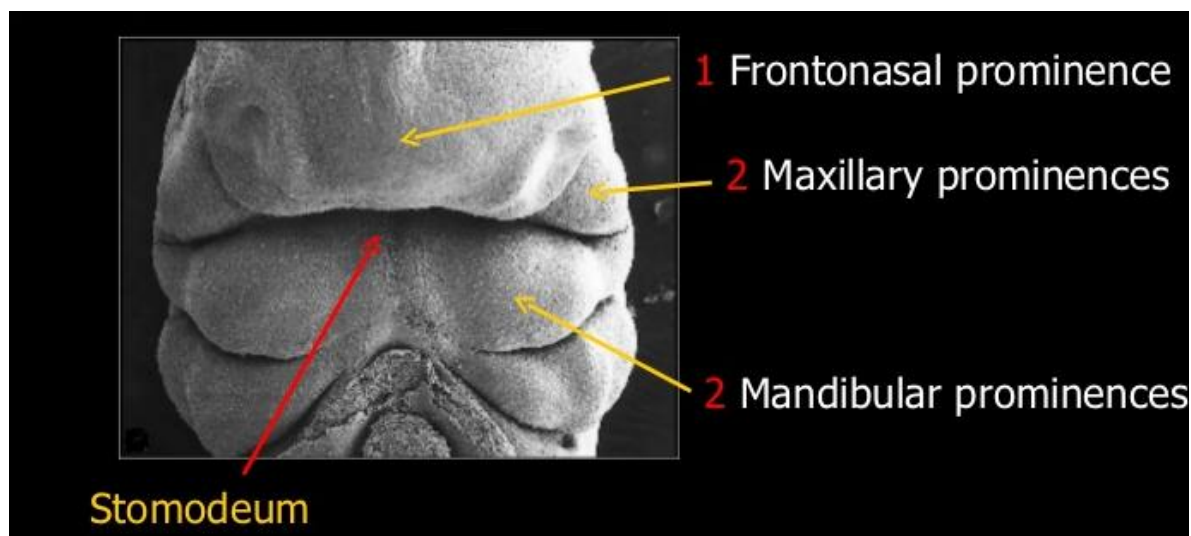
**Dr.Thuraya Milad Elhady**

## DEVELOPMENT OF THE FACE

The face develops from five primordial prominences that surround a central depression (the primitive oral pit) .

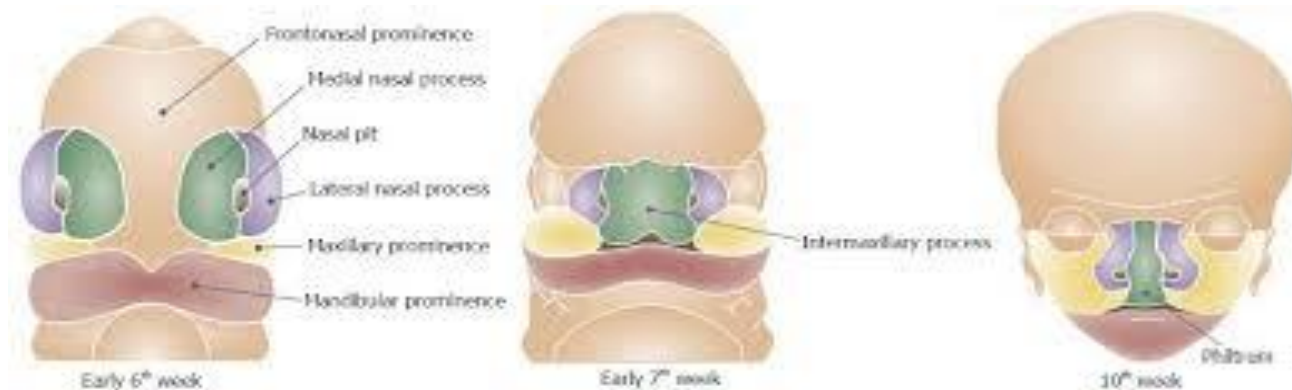
**At the fourth week** the ectomesenchymal cells migrate in two streams.

A cranial stream that causes mesodermal enlargement at the frontal area to form the frontonasal process ,and a caudal stream that enters the first pharyngeal arches and form two bilaterally located maxillary and mandibular processes . Thus the five primordial of the face are established .



The mandibular processes appear initially as bilateral structures but soon merge with each other at the midline , eliminating the depression or the groove between them .

These processes will give rise to the mandible , the lower part of the face (lower lip & lower cheek) .

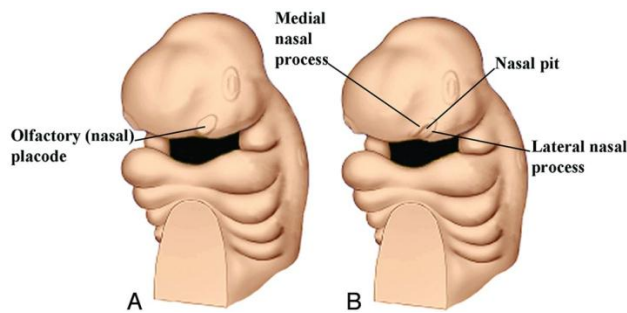


### Developmental Anomalies :

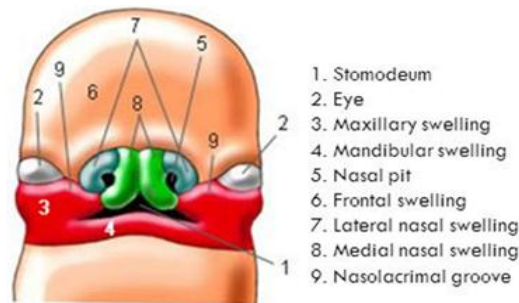
Median cleft of the mandible . This is a rare condition which results from failure of the mesenchymal masses of the mandibular processes to merge together at five weeks I.U. The mandible may then develop with no bony union in the midline . A dimple of the chin is the slightest form of incomplete merging of the two mandibular processes .



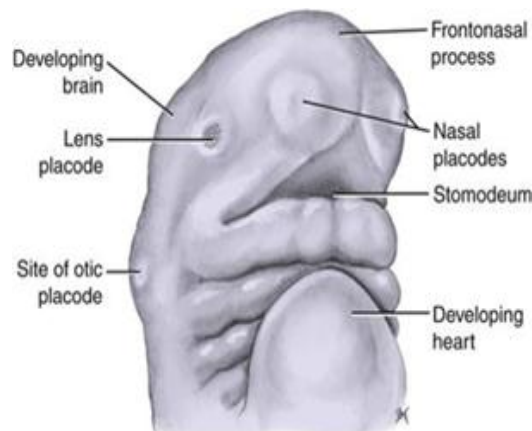
**In the fifth week :** The nasal placodes ( olfactory placodes ) develop bilaterally on the inferior lateral part of the frontonasal process where it borders the oral cavity . At the margins of the nasal placodes . the mesenchyme proliferates rapidly transforming the nasal placodes into nasal pits . The mesodermal proliferation is in the form of two horseshoe shaped elevations .The long arms of the horseshoes form the medial nasal processes and the short arms form the lateral nasal processes . The region of the frontal process where these changes take place and the nose will develop is called the frontonasal process .



The grooves between the median nasal processes and maxillary processes are called the lateral labial grooves or bucconasal grooves and the grooves between the lateral nasal processes and the maxillary processes are called the naso-optic grooves or naso-lacrimal groove .



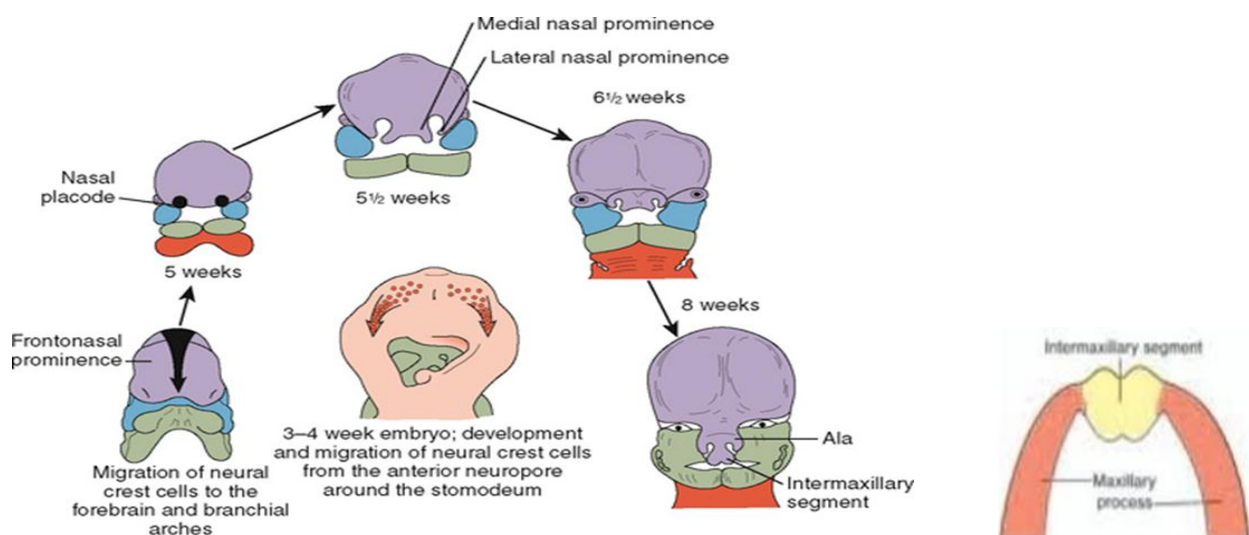
The eyes develop during the 5<sup>th</sup> week . The first external sign of eye development is the appearance of the lens placodes between the maxillary and the frontonasal processes at the lateral sides of the face .



**By the sixth week** , the mesoderm of maxillary processes proliferates rapidly causing the eyes to enlarge and grow medially . This medial migration of the maxillary processes moves the medial nasal processes toward the midline and toward each other . At the same time the mesoderm around the nasal pits continues its proliferation and enlarges bringing the medial nasal processes more anteriorly . The maxillary processes now approach the medial nasal processes but remain separated from them by the bucconasal grooves on both sides .

The maxillary and lateral nasal processes are initially separated by a deep furrow or groove called naso-lacrimal groove .The epithelium in the floor of this groove forms a solid cord of epithelial cells , and is eventually canalized to form the naso-lacrimal duct .Later , this duct becomes invested by bone during the ossification of the maxilla. After birth it functions to drain excess tears from the eye to the nasal cavity .

At the middle of the sixth week ,the two medial nasal processes merge in the midline to form the inter-maxillary segment .This will give rise to the central part of the nose including the nasal septum , the center of the upper lip (the philtrum and the primary palate ).\*The primary palate is that part of the alveolar process carrying the incisor teeth.



## **DEVELOPMENT OF FACIAL FEATURES**

The facial features develop at the fetal period. By the seventh week, the face has acquired a more human appearance. The facial features are accomplished by the differential growth of the different facial processes.

There is a decreased growth in breadth of the medial and lateral nasal processes and an increased growth of the maxillary processes. So, the eyes move from the lateral aspects of the head, into a position on either side of the nose. The orbital cavities attain their adult dimensions at about seven years of age.

The medial part of the face now increases in an anterior direction and at the same time the vertical height increases.

The face grows in a forward direction to form the nasal bridge so that the nostrils and eyes are not on the same plane. The nose is not fully developed in the newborn infant, as a baby has a flat nose. The nose acquires its inherited size and shape at puberty.

The vertical growth of the face moves the ears cranially from the lower part of the face to be positioned laterally in the middle third of the face.

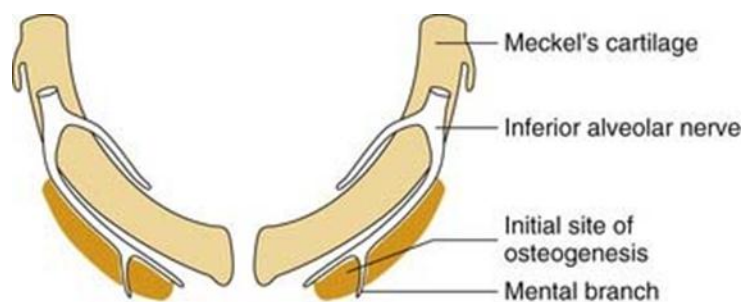
During early development until 8 weeks the mandible is small relative to upper face, and the lower lip is positioned behind the upper one. By 9 weeks the head is raised up and is lifted away from the thorax and thus permits the mandible to grow forward.

### **Development and ossification of the mandible**

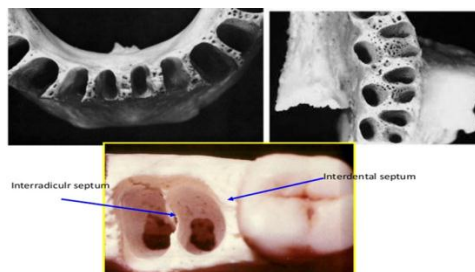
The mandible develops as an intramembranous bone lateral to Meckel's cartilage (a typical hyaline cartilage covered by fibrous tissue). It extends as a curved rod from the cartilaginous otic capsule at the base of the skull to the midline; where its ventral end turns upward in contact with the cartilage of the opposite side, to which it is joined by mesenchyme.

The dorsal end of Meckel's cartilage ossifies to form the malleus and incus of the inner ear . The ventral end at the symphysis forms accessory endochondral ossicles in the chin region . the rest of the cartilage disappears without contributing to the formation of the bone of the mandible leaving the mylohyoid groove in which it lays .the fibrous sheath surrounding the cartilage remains as the sphenomandibular and the sphenomalleolar ligaments .

Ossification of the mandible begins during the 6th week I. U . in the fibrous tissue lateral to Meckel's cartilage , by the appearance of single ossification center for each half of the mandible in the region of the bifurcation of the inferior alveolar nerve into mental and incisive branches ; that is in the region of the future mental foramen . from this center bone formation spreads rapidly to embrace the inferior alveolar nerve and to form the body proper of the mandible.



It extends in the anterior posterior direction from the symphysis to the mandibular foramen . The alveolar process of the mandible develop by the upward growth of the outer and inner plates of mandibular bone thus the tooth germ are contained in a groove of bone which opens towards the surface of the oral cavity . this groove is later divided by the development of bony septa between the tooth germs .

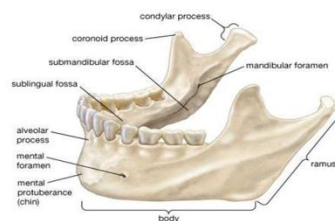


The backward extension of the mandible to form the ramus is produced by spread of ossification from the body ,behind and above the mandibular foramen mapping the ramus and its two processes . At the 10<sup>th</sup> and 16<sup>th</sup> W.I.U. , further growth of these processes occurs by the appearance of accessory cartilages forming secondary centers at the condyle and coronoid processes . A third accessory cartilage , the symphyseal cartilage or cartilage appears at the symphysis .

The condylar cartilage forms most of the developing ramus and condyle and acts as the very important growth center of the mandible up to 21 years of age .

The coronoid cartilage forms the coronoid process and disappears long before birth (transient growth cartilage ).

The symphyseal cartilage or cartilages appear in the connective tissue between the ends of Meckel's cartilage ( independent of it )and become ossified within the first year after the birth .



**Growth of the mandible :**the main sites of postnatal mandibular growth are at the condylar cartilage, the posterior border of the ramus and the alveolar ridge . These areas of bone apposition for the increased in the height , length and width of the mandible . They are however , accompanied by selective resorption which results in modeling and remodeling of the growing bone .

The mandible grows in three planes,vertically ,anteroposteriorly and trnsversally

**1.Vertical Growth :** is the result of

**a)Growth at the condyle .**

**b)Vertical growth of the body.**

a). **growth at the condyle** : the condyle cartilage serves as an important center of growth , for the ramus and the body of the mandible . The continuous proliferation of this cartilage until the age of 21 years produces an increase in mandibular length as measured from condyle to symphysis and at the same time an increase in the height of the ramus . Growth at the condyle usually occurs in a backward upward and outward direction , thus increasing the length of each half of the mandible .

The upward and backward growth of the condyle leads to moving the mandible downward and forward , thus providing the space into which the growth of the mandibular and maxillary alveolar processes can occur.

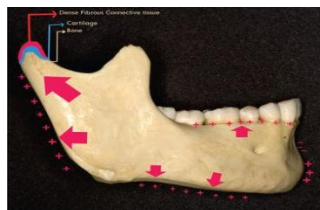
While the outward growth of the condyles increase the distance between the two condyles to accommodate the increasing width of the base of the skull .

b.**Vertical growth of the body of the mandible** : This occurs by surface apposition of bone and results in an increase in height of the mandibular body .

2.**Antero-posterior growth** : bone apposition occurs on the posterior border of the ramus . This is accompanied with resorption on the anterior border to maintain the proportion of the ramus and to give room for the eruption fo the subsequent molars particularly the permanent ones .

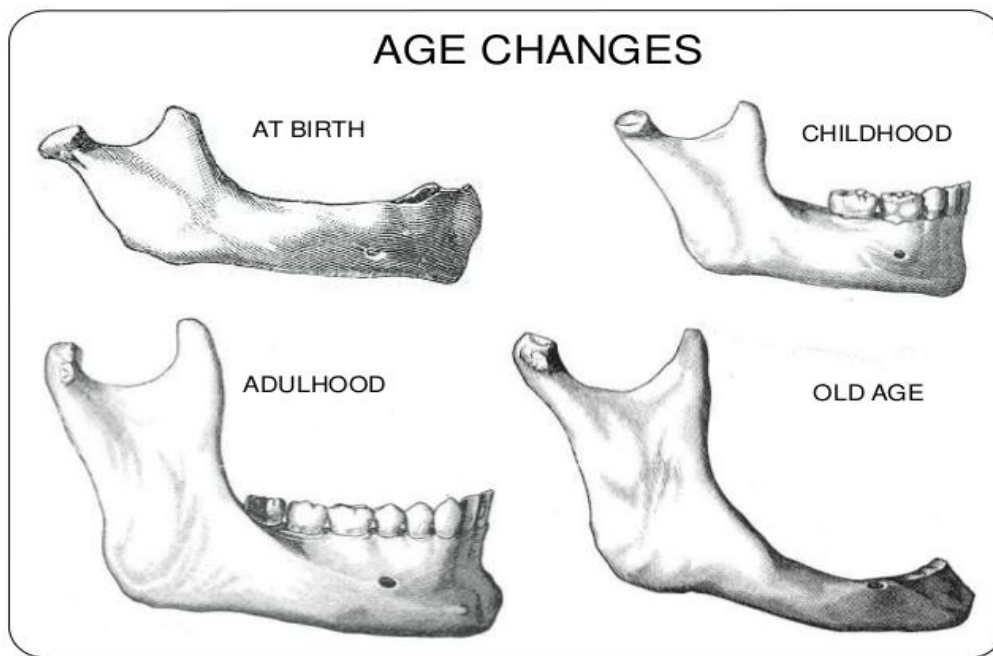
3.**Transverse growth** : The growth here occurs by periosteal apposition on the external surface of the mandible which results in an increase in thickness of the mandible .

This periosteal apposition of bone is accompanied by internal resorption and reconstruction of bone .



## Age changes of the mandible :

- ✓ A birth the mandible differs from adult bone.
  - the mandibular angle between the lower border and the posterior border of the ramus is about  $170^{\circ}$  at birth
  - the ramus is small and short and the coronoid process is higher than the condyle process .
  - The chin is usually poorly developed and the two halves of the mandible are joined by fibrous tissue in the midline .
  - The mental foramen is near the lower border below the crypt of the first deciduous molar and the tooth germs lie in a groove of bone with no bone overlying them at this stage .
- ✓ The growth and modeling changes occurring during infancy and childhood serve to alter the mandibular angle from about  $140^{\circ}$  in childhood to about  $110^{\circ}$  in adult .
- ✓ The condylar process becomes higher than the coronoid process .
- ✓ In senility the resorption of the alveolar bone with the loss of teeth results in increasing the angle and in backward inclination of the condyle .

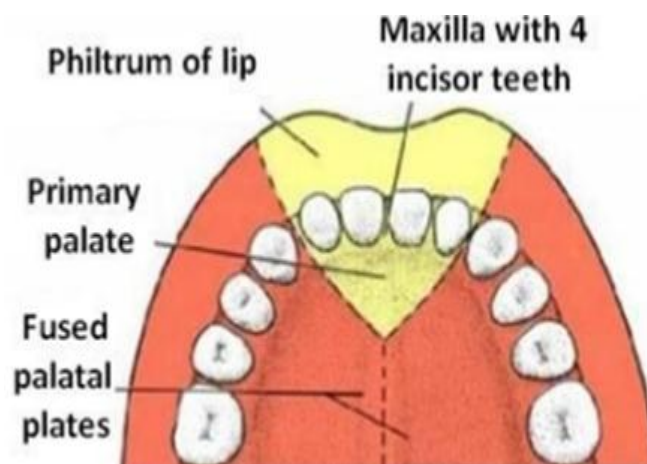


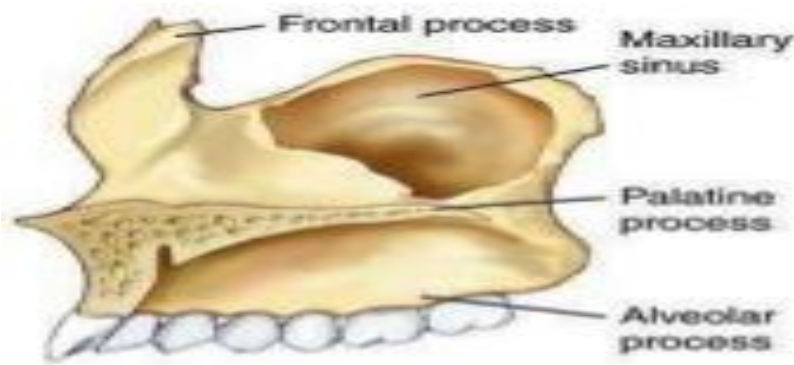
## **Location of mental foramen from birth to old age :**

1. At birth and early childhood , the mental foramen is very near to the lower border of the mandible .
2. In childhood , after eruption of the deciduous set the foramen is just above the midway line .
3. In adult , when all the permanent teeth are in position , the foramen is just below the midway line . This is because of the higher permanent sockets .
4. In old age , after the loss of teeth , the foramen is very near to the upper border . This is because of the resorption and nearly complete disappearance of the alveolar process .

## **❖ DEVELOPMENT AND OSSIFICATION OF THE MAXILLA**

The human maxilla is composed of two bones , the maxilla proper and the premaxilla .In adult no line of demarcation is seen between them .However , this union is indicated on the palatal aspects of young skulls by the incisive fissure extending from the incisive foramen to the alveolus of the canine .On the facial aspects all trace of union between maxilla and premaxilla disappears before birth.





Ossification in the maxilla commences at the 7th W.I.U. , by the appearance of one center of ossification close to the anterosuperior dental nerve ( at the future canine fossa ). From this center ossification spreads upward, downward , backward ,inward and anterior to form frontal,the alveolar , the zygomatic and the palatine processes respectively and to spread the ossification anteriorly , to the premaxilla .



There are two premaxilla , one on either side of the midline of the face . they develop in the mesenchyme of the median nasal processes .

Each premaxilla is a piece of bone which supports the maxillary incisor teeth and forms the anterior part of the hard palate up to the incisive foramen .

A secondary cartilage ( zygomatic or malar )appears in the developing zygomatic process , and contributes to the development of the maxilla .

### **\*\*\*Growth of maxilla : they are 2 types**

#### **1.Growth at the suture .      2. Growth by surface apposition .**

##### **1.Growth at the sutures :**

During the first few years after birth , there is certain amount of growth at the sutures between the maxilla and the other facial bones ( palatine , nasal, lacrimal and zygomatic )

##### **2.Growth by the surface apposition :**

It occurs by periosteal apposition of bone on the outer surface of the maxilla, on the posterior surface of the maxillary tuberosity , the alveolar process and the oral surface of the hard palate .

This periosteal apposition is compensated for by bone resorption in the interior of the maxilla and in the nasal surface of the hard palate thereby increasing the size of the sinuses and the depth of the nasal cavities .

In general the balance between osteoblastic apposition and osteoclastic resorption has a large effect on determining the final shape and size of the bones of the face .

### **Development of lips**

#### **The lower lip develops during the fourth or fifth week I.U. by merging of the mandibular processes .**

The upper lip develops from the maxillary processes and the medial nasal processes . It begins to develop at the sixth week I.U. when the maxillary processes approach the medial nasal processes but remain separated from them by the bucconasal grooves .

By the middle of the sixth week , the two medial nasal processes merge in the midline to form the inter-maxillary segment (globular process) . This will give rise to the central part of the upper lip (the philtrum) .

At the seventh week merging takes place between the two maxillary processes and the lateral boundaries of the intermaxillary segment eliminating the bucconasal grooves in between . **\*So, the upper lip is formed from one third medial nasal processes and two thirds maxillary processes .**

At the same time the mesoderm of the first pharyngeal arches with its sensory innervations from the maxillary divisions of the trigeminal nerve invade the upper lip and cheeks, while the mesoderm of the first pharyngeal arches with its mandibular divisions of the trigeminal nerve invades the lower lip .

### **Developmental Anomalies**

**CLEFT LIP :** This is a malformation of the upper lip .It may be unilateral or bilateral . The cleft varies from a notch in the vermilion border to a cleft extending into the floor of the nostril and through the alveolar ridges .

**Unilateral cleft lip :** Unilateral cleft lip results from failure of the maxillary process on one side to meet and fuse with the intermaxillary segment. This results in the persistence of the labial groove which is stretched and breaks down leading to the division of the lip into medial and lateral parts .

Unilateral cleft lip results in nasal distortion as the lip and nasal tissues are pulled toward the attached side .



**Bilateral cleft lip :** results from failure of the maxillary processes on both sides to meet and fuse with the intermaxillary segment . The defect may be symmetrical or asymmetrical .



**Median cleft lip :** is extremely rare and results from partial or complete failure of the medial nasal processes to merge together . This is the only type of cleft lip that may be called a “hare lip “. Rabbits normally have a cleft in the middle of the lip and this is where this term came from .



***Good luck***